

Evidence-Based Diet for Male Infertility & Obesity

(Body Mass Index ≥ 28)

A research synthesis of dietary patterns, food-by-food nutritional values, and the mechanisms by which specific foods support sperm quality, hormonal balance, metabolic health, and weight loss.

Compiled from peer-reviewed systematic reviews, meta-analyses, and randomized controlled trials indexed in PubMed/MEDLINE.

Audience:Adult men with sub-fertility and/or BMI 28+

Document type:Educational synthesis (not medical advice)

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Important disclaimer

This document is a structured summary of the published research literature. It is intended for educational use only and is NOT a substitute for personalized advice from a licensed urologist, reproductive endocrinologist, internist, or registered dietitian. Caloric targets, macronutrient ratios, and supplement doses should be tailored to the individual after appropriate clinical evaluation (semen analysis, hormone panel, metabolic labs, comorbidities, allergies, and current medications). Some supplements (e.g., high-dose selenium) are harmful in excess.

1. Executive summary

Two clinical problems are addressed jointly in this document because they are biologically coupled: excess adiposity is one of the most common modifiable causes of poor semen quality and low testosterone, and the same dietary pattern that produces durable weight loss in BMI ≥ 28 also improves sperm parameters.

Key conclusions from the literature

- A Mediterranean-style diet, with optional DASH features, has the strongest and most consistent evidence base for both weight loss in adults with BMI ≥ 25 and improvement of conventional semen parameters (concentration, total motility, normal morphology).
- Even modest weight loss (5-10% of body weight) in obese men is associated with measurable increases in serum total testosterone, SHBG, and sperm concentration, and reductions in estradiol and sperm DNA fragmentation.
- Specific foods - oily fish, walnuts and other tree nuts, leafy greens, tomatoes, pomegranate, eggs, legumes, oats, and extra virgin olive oil - deliver the bioactive compounds (omega-3, folate, zinc, selenium, lycopene, polyphenols, vitamin E, L-carnitine, CoQ10, vitamin D) that drive these benefits.
- Western dietary patterns (ultra-processed foods, sugar-sweetened beverages, processed red meats, trans fats, high refined-carb load) are consistently associated with worse semen parameters and weight gain.
- A practical 12-week program combining a 500 kcal/day energy deficit, ~ 1.4 g/kg target-body-weight protein, Mediterranean food choices, 150 minutes/week of moderate aerobic activity, and 2 resistance sessions/week is the recommended starting intervention.

Targets used in this document

Energy intake (BMI 28-30) 1700-2000 kcal/day (500 kcal deficit from TDEE)

Energy intake (BMI 30+) 1500-1800 kcal/day, reassess every 4 weeks

Protein: 1.2-1.6 g per kg target body weight per day

Fat: 30-35% of energy, mostly mono- and polyunsaturated

Carbohydrate: 40-50% of energy, predominantly low glycemic index

Fiber: ≥ 30 g/day

Added sugar: ≤ 25 g/day

Alcohol: ≤ 7 standard drinks/week, ideally none during preconception

2. Why obesity and male infertility share a diet

2.1 Sperm parameters that respond to diet

A standard semen analysis reports volume, concentration, total motility, progressive motility, and morphology. Advanced tests add DNA fragmentation index (DFI) and reactive oxygen species (ROS). Diet influences each of these:

- Concentration and total count: respond to overall energy balance, trans fat avoidance, and adequate folate, zinc, and vitamin D.
- Motility: responds strongly to omega-3 (DHA), L-carnitine, coenzyme Q10, and selenium.
- Morphology: responds to antioxidant intake (vitamin C, E, lycopene, polyphenols) and to walnuts/tree nut intake.
- DNA fragmentation: responds to antioxidants and to a reduction in oxidative load (reduced ultra-processed food, alcohol, smoking).

2.2 The obesity link

Excess visceral adipose tissue increases peripheral aromatization of testosterone to estradiol, lowers SHBG, drives chronic low-grade inflammation, raises scrotal temperature via suprapubic fat, and elevates seminal oxidative stress. Meta-analyses report that men with BMI ≥ 30 have lower sperm concentration, total motility, and normal morphology, and higher DNA fragmentation than normal-weight men. Importantly, these changes are at least partially reversible with weight loss.

2.3 What 'BMI 28+' means in practice

BMI 25.0-29.9 is classified as overweight; BMI ≥ 30 is obese. A man at BMI 28 is in the upper overweight range with elevated metabolic and reproductive risk. The intervention threshold and the diet structure are essentially identical from BMI 28 upward; only the energy deficit, monitoring intensity, and likely duration of treatment differ.

3. Dietary patterns with the strongest evidence

3.1 Mediterranean diet (foundation pattern)

Defined by high intake of vegetables, fruit, whole grains, legumes, nuts, olive oil, and fish; moderate intake of poultry, eggs, dairy (especially yogurt and fresh cheese); low intake of red meat, processed meat, and added sugar. Multiple meta-analyses show this pattern is associated with weight loss in adults with overweight or obesity and with improved conventional semen parameters in men.

Operational rules used in this plan

- ≥ 5 servings of vegetables and fruit per day (mostly vegetables).
- ≥ 3 servings of legumes per week.
- ≥ 30 g of mixed nuts per day (walnuts, almonds, hazelnuts).
- Extra virgin olive oil as the main added fat (≥ 2 tablespoons/day).
- Oily fish 2-3 times per week (salmon, sardines, mackerel, anchovies).
- Whole grains instead of refined (oats, barley, bulgur, whole rye).
- Red meat ≤ 1 serving/week; processed meat avoided.
- Sugar-sweetened beverages avoided.

3.2 Higher-protein modification for weight loss

For men with BMI ≥ 28 the basic Mediterranean template is modified in two ways: total energy is reduced by ~ 500 kcal/day below estimated total daily energy expenditure, and protein is raised to 1.2-1.6 g/kg of target body weight (preserves lean mass during deficit, increases satiety, and elevates the thermic effect of feeding). Practically: every meal contains a deliberate protein anchor (eggs, Greek yogurt, fish, lentils, tofu, skinless poultry, low-fat dairy, whey).

3.3 DASH features layered on top

If blood pressure is elevated, the DASH (Dietary Approaches to Stop Hypertension) emphasis on potassium-rich produce and reduced sodium (< 2 g/day) is layered on. DASH and Mediterranean overlap heavily.

3.4 Patterns to avoid

- 'Western' pattern: processed meat, refined grains, sugary drinks, fried foods, ultra-processed snacks. Associated with lower sperm count and motility and with weight gain.
- Very-low-carbohydrate (ketogenic) for general weight loss is effective short term but evidence for fertility benefit is weaker than Mediterranean; sustainability and lipid profile responses vary.
- Liquid meal-replacement-only protocols can produce rapid weight loss but, without dietary education, weight regain is common; not preferred for the preconception period.

4. Food monographs

Each card lists the food, an approximate macro/micronutrient profile per 100 g of edible portion (rounded; from USDA FoodData Central and EuroFIR-aligned national databases), the bioactive compounds that matter for fertility and weight management, and the mechanisms by which the food acts. Serving sizes given are practical, not strict.

Walnuts (*Juglans regia*)

30 g serving (~14 halves)

Nutrition per 100 g 654 kcal; protein 15 g; fat 65 g (of which alpha-linolenic acid (ALA, omega-3) ~9 g, linoleic acid ~38 g, monounsaturated ~9 g); carbohydrate 14 g; fiber 6.7 g; magnesium 158 mg; phosphorus 346 mg; zinc 3.1 mg; copper 1.6 mg; manganese 3.4 mg; folate 98 microg; vitamin E 0.7 mg.

Key bioactives Alpha-linolenic acid (plant omega-3), polyphenols (ellagitannins, urolithins after gut metabolism), gamma-tocopherol, L-arginine, melatonin.

Mechanism / benefit RCTs (Robbins 2012; Salas-Huetos 2018) showed 75 g/day of walnuts for 12 weeks improved sperm vitality, motility, morphology, and reduced sperm aneuploidy in men on Western diets. Mechanisms: ALA-derived DHA enrichment of sperm membranes, polyphenol-driven reduction of seminal oxidative stress, L-arginine support of nitric-oxide-mediated vasodilation in the testis, and zinc/copper cofactor delivery.

Practical use 30 g/day as snack or salad/oats topping. Replace, do not add, calorie-equivalent ultra-processed snacks.

Salmon (wild or responsibly farmed Atlantic)

150 g cooked portion 2-3x/week

Nutrition per 100 g Per 100 g cooked: 208 kcal; protein 22 g; fat 13 g (EPA ~0.7 g, DHA ~1.4 g); vitamin D 526 IU; vitamin B12 3.2 microg; selenium 36 microg; potassium 363 mg; phosphorus 252 mg; choline 95 mg.

Key bioactives Long-chain omega-3 EPA and DHA, vitamin D, selenium, astaxanthin, B12, choline.

Mechanism / benefit DHA is incorporated into sperm tail membranes and is required for motility and acrosome integrity; EPA reduces systemic inflammation. Vitamin D supports Leydig-cell testosterone production. Selenium is a cofactor for glutathione peroxidase, the principal sperm antioxidant defense. Combined effect: improved motility, reduced DNA fragmentation, support of testosterone.

Practical use Baked, grilled, or pan-seared 2-3 times per week. Sardines and mackerel deliver similar omega-3 at lower cost; anchovies are a high-yield option.

Pumpkin seeds (*Cucurbita pepo*, kernels)

30 g handful

Nutrition per 100 g Per 100 g: 559 kcal; protein 30 g; fat 49 g (mostly linoleic and oleic acid); carbohydrate 11 g; fiber 6 g; magnesium 592 mg; phosphorus 1233 mg; zinc 7.8 mg; iron 8.8 mg; manganese 4.5 mg; potassium 809 mg.

Key bioactives Zinc, magnesium, plant sterols, L-arginine, tryptophan, gamma-tocopherol.

Mechanism / benefit Zinc is concentrated in seminal plasma and is required for spermatogenesis, sperm membrane stability, and testosterone synthesis. Magnesium supports insulin sensitivity and sleep quality. Plant sterols modestly reduce LDL cholesterol.

Practical use 30 g/day as snack, on yogurt, or stirred into salads. Avoid heavily salted or candied versions.

Brazil nuts (*Bertholletia excelsa*)

1-2 nuts/day - do not exceed

Nutrition per 100 g Per 100 g: 659 kcal; protein 14 g; fat 67 g; carbohydrate 12 g; fiber 7.5 g; selenium 1917 microg (varies widely); magnesium 376 mg; thiamine 0.6 mg.

Key bioactives Selenium (the densest dietary source on earth), tocopherols, magnesium.

Mechanism / benefit Selenium incorporates into selenoproteins including glutathione peroxidase 4 (GPx4), which is essential for sperm structural integrity. Even 1-2 Brazil nuts/day raises plasma selenium into the optimal range for sperm motility. Crucially, selenium toxicity occurs above ~400 microg/day, so 1-2 nuts is the ceiling, not a target.

Practical use 1 nut/day if other selenium sources are present; 2 nuts/day if not. Store in cool dark conditions.

Eggs (whole, large, ~50 g each)

2-3 per day acceptable in this pattern

Nutrition per 100 g Per 100 g (~2 large): 143 kcal; protein 12.6 g; fat 9.5 g (saturated 3.1 g); cholesterol 372 mg; vitamin A 540 IU; vitamin D 82 IU; vitamin B12 0.9 microg; choline 294 mg; selenium 30 microg; lutein + zeaxanthin 503 microg; iodine ~25 microg.

Key bioactives High-biological-value protein, choline, vitamin D, B12, lutein/zeaxanthin, selenium, iodine.

Mechanism / benefit Choline is required for one-carbon metabolism and sperm membrane phosphatidylcholine. Eggs are one of the few dietary vitamin D sources outside oily fish. High-quality protein (DIAAS ~1.13) supports satiety in a calorie deficit. Current evidence does not link normal egg intake (up to ~1/day in most cohorts; up to 2-3/day in metabolically healthy men) with cardiovascular harm.

Practical use Boiled, poached, or scrambled in olive oil. Pair with vegetables and a complex carbohydrate.

Greek yogurt (plain, 0-2% fat)

200 g per serving

Nutrition per 100 g Per 100 g (0% fat): 59 kcal; protein 10 g; fat 0.4 g; carbohydrate 3.6 g; calcium 110 mg; phosphorus 135 mg; potassium 141 mg; vitamin B12 0.8 microg.

Key bioactives Whey- and casein-derived branched-chain amino acids, conjugated linoleic acid (in fuller-fat versions), probiotic cultures (*Lactobacillus*, *Streptococcus thermophilus*).

Mechanism / benefit Dense, satiating protein source for use in calorie deficits; supports lean mass preservation and muscle protein synthesis. Probiotic cultures may modestly improve gut barrier function and inflammation profile. Calcium contributes to blood pressure regulation.

Practical use 200 g/day as breakfast or snack base; pair with berries, nuts, oats. Avoid pre-sweetened flavored varieties.

Lentils (red, green, brown, cooked)

1 cup cooked (~200 g)

Nutrition per 100 g Per 100 g cooked: 116 kcal; protein 9 g; fat 0.4 g; carbohydrate 20 g; fiber 8 g; folate 181 microg; iron 3.3 mg; magnesium 36 mg; potassium 369 mg; zinc 1.3 mg.

Key bioactives Folate, soluble + insoluble fiber, plant protein, polyphenols (catechins, procyanidins).

Mechanism / benefit Folate is a one-carbon donor required for spermatogenesis and DNA synthesis; men in higher folate quartiles have lower rates of sperm aneuploidy. High fiber improves glycemic control and satiety, supporting weight loss. Plant protein replaces processed meat in this pattern.

Practical use 3+ servings/week. Use as base for soups, stews, or grain bowls. Soaking and rinsing reduces phytate and gas.

Chickpeas (*Cicer arietinum*, cooked)

1 cup cooked (~165 g)

Nutrition per 100 g Per 100 g cooked: 164 kcal; protein 8.9 g; fat 2.6 g; carbohydrate 27 g; fiber 7.6 g; fat 2.6 g; carbohydrate 27 g; fiber 7.6 g; folate 172 microg; iron 2.9 mg; magnesium 48 mg; manganese 1 mg; phosphorus 168 mg; zinc 1.5 mg.

Key bioactives Folate, manganese, resistant starch, saponins, polyphenols.

Mechanism / benefit Resistant starch acts as a prebiotic, generating short-chain fatty acids that improve insulin sensitivity. Manganese is a cofactor for superoxide dismutase, an antioxidant relevant to sperm.

Practical use Hummus, salads, stews. 3+ servings/week.

Spinach (raw or lightly cooked)

150 g cooked or 250 g raw

Nutrition per 100 g Per 100 g raw: 23 kcal; protein 2.9 g; fat 0.4 g; carbohydrate 3.6 g; fiber 2.2 g; folate 194 microg; vitamin K 483 microg; vitamin A 9377 IU; vitamin C 28 mg; iron 2.7 mg; magnesium 79 mg; potassium 558 mg; nitrates ~0.7 g/kg.

Key bioactives Folate, beta-carotene, lutein/zeaxanthin, vitamin C, dietary nitrates, magnesium.

Mechanism / benefit Folate-rich; nitrates convert to nitric oxide, improving endothelial function and erectile/penile blood flow. Magnesium supports insulin sensitivity. High micronutrient density per kcal supports a calorie deficit.

Practical use Daily; lightly cooked spinach has higher iron bioavailability per serving than raw.

Broccoli

200 g cooked

Nutrition per 100 g Per 100 g cooked: 35 kcal; protein 2.4 g; fat 0.4 g; carbohydrate 7.2 g; fiber 3.3 g; fat 0.4 g; carbohydrate 7.2 g; fiber 3.3 g; vitamin C 65 mg; folate 108 microg; vitamin K 141 microg; potassium 293 mg; calcium 40 mg.

Key bioactives Sulforaphane, indole-3-carbinol, vitamin C, folate.

Mechanism / benefit Sulforaphane activates the Nrf2 antioxidant pathway, reducing oxidative stress relevant to sperm DNA integrity. Indole-3-carbinol modulates estrogen metabolism toward less mitogenic forms. Volume-dense and low-energy: helps with satiety in a calorie deficit.

Practical use Steam or roast, do not over-cook (preserves sulforaphane). Pair with olive oil for vitamin K absorption.

Tomatoes (especially cooked / paste)

200 g raw or 50 g paste

Nutrition per 100 g Per 100 g raw: 18 kcal; protein 0.9 g; fat 0.2 g; carbohydrate 3.9 g; fiber 1.2 g; vitamin C 13.7 mg; potassium 237 mg; lycopene ~3 mg; folate 15 microg. Tomato paste (100 g): 82 kcal; lycopene up to 30 mg.

Key bioactives Lycopene (highest bioavailability after cooking with oil), beta-carotene, vitamin C, potassium.

Mechanism / benefit Lycopene supplementation has been associated with improved sperm motility, morphology, and concentration in meta-analyses of RCTs. As a singlet-oxygen quencher it reduces seminal oxidative stress; it also accumulates in the testis and prostate.

Practical use Daily. Cook with olive oil to maximize lycopene absorption (e.g., a Mediterranean tomato sauce).

Pomegranate (arils and 100% juice)

150 g arils or 200 mL juice

Nutrition per 100 g Per 100 g arils: 83 kcal; protein 1.7 g; fat 1.2 g; carbohydrate 19 g; fiber 4 g; vitamin C 10 mg; vitamin K 16 microg; folate 38 microg; potassium 236 mg.

Key bioactives Punicalagins and ellagitannins, anthocyanins, ellagic acid (gut metabolites: urolithins).

Mechanism / benefit Animal and small human RCT data report increased sperm motility and

reduced lipid peroxidation in seminal plasma. Mechanism: potent antioxidant and anti-inflammatory activity, with mitochondrial-supportive urolithin A acting on sperm energetics.

Practical use 150 g arils as snack or salad addition; 200 mL unsweetened juice no more than once daily (sugar load).

Berries (blueberry, strawberry, raspberry, blackberry)

100-150 g/day

Nutrition per 100 g Per 100 g (mixed average): 50 kcal; protein 1 g; carbohydrate 12 g; fiber 3-6 g; vitamin C 10-60 mg; manganese 0.3-0.6 mg; folate 6-25 microg; total polyphenols 200-700 mg.

Key bioactives Anthocyanins, flavonols (quercetin), ellagic acid, vitamin C, fiber.

Mechanism / benefit Anthocyanins reduce oxidative stress and improve endothelial function. Low glycemic index helps glycemic control. High satiety per kcal helps weight loss adherence.

Practical use Daily in oats, yogurt, or as a snack. Frozen berries are nutritionally equivalent and cheaper.

Oats (*Avena sativa*, rolled or steel-cut)

60 g dry / day

Nutrition per 100 g Per 100 g dry: 379 kcal; protein 13.2 g; fat 6.5 g; carbohydrate 67 g; fiber 10 g (of which beta-glucan ~4 g); magnesium 138 mg; iron 4.7 mg; zinc 4 mg; manganese 3.6 mg; phosphorus 410 mg.

Key bioactives Beta-glucan (soluble fiber), avenanthramides (oat-specific polyphenols), magnesium.

Mechanism / benefit Beta-glucan slows gastric emptying, blunts postprandial glucose excursions, and lowers LDL cholesterol via bile acid binding. Avenanthramides have anti-inflammatory and antioxidant activity. Together: better satiety in a calorie deficit and better cardiometabolic profile.

Practical use Daily breakfast base; combine with milk, walnuts, berries.

Extra virgin olive oil (Mediterranean primary fat)

2-3 tbsp/day (30-45 mL)

Nutrition per 100 g Per 100 g: 884 kcal; fat 100 g (oleic acid 73 g, palmitic 11 g, linoleic 7 g); vitamin E 14 mg; vitamin K 60 microg; polyphenols 50-800 mg/kg in EVOO.

Key bioactives Oleic acid, oleocanthal, oleuropein, hydroxytyrosol, vitamin E.

Mechanism / benefit Oleocanthal is a natural ibuprofen-like COX inhibitor; olive polyphenols reduce LDL oxidation and improve endothelial function. Replacing butter, palm oil, or refined seed oils with EVOO is one of the most reliable single dietary changes for cardiometabolic risk.

Practical use Use cold for dressings and finishing; sufficient smoke point for moderate-heat cooking (180-190 C).

Avocado (*Persea americana*)

1/2 fruit (~100 g)

Nutrition per 100 g Per 100 g: 160 kcal; protein 2 g; fat 14.7 g (oleic acid 9.8 g, palmitic 2.1 g); carbohydrate 8.5 g; fiber 6.7 g; potassium 485 mg; folate 81 microg; vitamin E 2.1 mg; vitamin K 21 microg; magnesium 29 mg.

Key bioactives Monounsaturated fat, soluble + insoluble fiber, lutein, beta-sitosterol.

Mechanism / benefit Improves HDL particle quality, supports satiety, displaces saturated-fat-heavy spreads. Beta-sitosterol modestly lowers LDL cholesterol.

Practical use 1/2 fruit/day as spread, salad, or topping. Watch energy density if total deficit is tight.

Green tea (*Camellia sinensis*)

2-3 cups/day

Nutrition per 100 g Per 240 mL brewed: ~2 kcal; caffeine 25-50 mg; EGCG 100-300 mg; theanine 25-60 mg; fluoride ~0.2 mg.

Key bioactives Epigallocatechin gallate (EGCG) and other catechins, L-theanine, caffeine.

Mechanism / benefit Catechins are potent antioxidants and have a small but reproducible thermogenic and lipid-oxidation effect; L-theanine smooths the caffeine response. Useful behavioral substitute for sweetened beverages.

Practical use 2-3 cups/day; avoid late-day intake if sleep is sensitive.

Dark chocolate (>= 85% cacao)

10-20 g/day (1-2 squares)

Nutrition per 100 g Per 100 g (85% cacao): 600 kcal; protein 7.8 g; fat 46 g; carbohydrate 24 g; fiber 11 g; iron 12 mg; magnesium 230 mg; copper 1.8 mg; flavanols ~300-700 mg.

Key bioactives Cocoa flavanols (epicatechin), magnesium, theobromine.

Mechanism / benefit Cocoa flavanols improve flow-mediated dilation (a marker of endothelial function), modestly lower blood pressure, and may improve insulin sensitivity. The energy density is high, so portion control matters.

Practical use 1-2 small squares as a planned dessert. Choose >= 85% cacao to limit sugar.

Whole grains (oats, barley, bulgur, whole rye, whole-wheat)

60-80 g dry/meal

Nutrition per 100 g Approximate per 100 g cooked: 110-150 kcal; protein 3-6 g; fiber 3-7 g; B vitamins; magnesium; iron; zinc.

Key bioactives Beta-glucan (oats, barley), arabinoxylans (rye, wheat), lignans, B vitamins, magnesium, zinc, selenium.

Mechanism / benefit Replacement of refined grains with whole grains lowers postprandial glucose and insulin spikes, supports gut microbiota diversity, and is associated in cohort studies with lower body weight gain over time and lower all-cause mortality.

Practical use Default starch source at all meals.

Tofu / tempeh / edamame (whole-food soy)

100-150 g/serving

Nutrition per 100 g Tofu firm per 100 g: 144 kcal; protein 17 g; fat 9 g; carbohydrate 3 g; calcium 350 mg (calcium-set); iron 2.7 mg; magnesium 58 mg; isoflavones 20-30 mg.

Key bioactives Soy protein, isoflavones (genistein, daidzein), calcium (in calcium-set tofu).

Mechanism / benefit Whole-food soy in normal dietary amounts has not been shown to reduce testosterone in healthy men; meta-analyses are reassuring. It is a useful low-saturated-fat protein replacement for processed meat.

Practical use 1-3 servings/week as part of legume rotation. Avoid high-dose isolated isoflavone supplements unless prescribed.

Skinless poultry (chicken or turkey breast)

120-150 g cooked

Nutrition per 100 g Per 100 g cooked breast: 165 kcal; protein 31 g; fat 3.6 g; selenium 27 microg; phosphorus 220 mg; B vitamins (niacin, B6).

Key bioactives High-biological-value protein, niacin, B6, selenium.

Mechanism / benefit Lean protein anchor for meals during a calorie deficit; preserves lean mass; high satiety per kcal.

Practical use Several times per week as lean protein. Bake, grill, or poach; avoid deep-frying or breaded preparations.

Garlic and onion (Allium spp.)

Daily, in cooking

Nutrition per 100 g Garlic per 100 g: 149 kcal; vitamin C 31 mg; manganese 1.7 mg; selenium 14 microg; allicin (formed from alliin when crushed).

Key bioactives Allicin, S-allyl cysteine, quercetin (onion), organosulfur compounds.

Mechanism / benefit Modest blood-pressure-lowering and lipid-lowering effects in meta-analyses; antioxidant and antimicrobial activities in vitro. Useful flavour foundation that allows reduction of salt and added fat.

Practical use Daily aromatic base for Mediterranean cooking.

Nutrition per 100 g 0 kcal.

Key bioactives Hydration; electrolyte balance with adequate dietary potassium and modest sodium.

Mechanism / benefit Adequate hydration supports semen volume and metabolic function. Replacing one daily 350 mL sugar-sweetened beverage with water is one of the largest single-leverage moves for weight loss in this BMI range.

Practical use Water, sparkling water, unsweetened tea, black coffee. Avoid sweetened beverages.

5. Foods and substances to limit or avoid

These are associated with worse semen parameters, weight gain, and/or metabolic deterioration. Frequency is the key concept: occasional intake is fine; daily/weekly habit is the problem.

Sugar-sweetened beverages

Sodas, energy drinks, sweetened juices. Each daily 250 mL serving is associated with measurable reductions in sperm motility and with weight gain in cohort studies.

Processed and cured meats

Bacon, sausage, hot dogs, deli meats. High in sodium, nitrites, saturated fat; associated with lower sperm morphology scores and higher cardiometabolic risk.

Trans-fat-containing foods

Industrially produced partially hydrogenated oils (legacy bakery items, some margarines, some fried fast food). Trans fats appear in seminal lipids and are associated with reduced total sperm count.

Deep-fried fast food

High energy density, oxidized fats, trans fats, low micronutrient yield. Strongly associated with the Western dietary pattern.

Refined grains and added sugar

White bread, white rice in excess, pastries, candy. High glycemic load worsens insulin sensitivity, which independently impairs Leydig cell function.

High-mercury fish

King mackerel, swordfish, shark, tilefish, big-eye tuna. Mercury is reproductively toxic; choose salmon, sardines, anchovies, trout, light canned tuna in moderation instead.

Excess alcohol

More than 7 standard drinks/week is dose-dependently associated with reduced sperm parameters and with hepatic and metabolic harm. During preconception, abstinence is the safest choice.

Soy in pharmacologic amounts

Whole-food soy in normal dietary amounts (tofu, edamame, tempeh) is fine and useful for protein. Very high isoflavone supplements are where caution is warranted.

Tobacco and recreational drugs

Not foods, but the highest-impact modifiable factors after weight: smoking, vaping nicotine, cannabis (especially daily), and anabolic steroids profoundly impair fertility.

6. Sample 7-day meal plan (~1800 kcal)

Calibrated for a man around 90 kg targeting ~5-7 kg loss over 12 weeks. Adjust portions of starch and oil up or down by ~15% to match the individual's TDEE. All days include $\geq$ 30 g fiber and $\geq$ 110 g protein.

Day 1 (Mon)

- Breakfast: Steel-cut oats 60 g (dry) cooked in 250 mL skim milk + 30 g walnuts + 100 g blueberries + 1 tsp cinnamon.
- Lunch: Grilled salmon 150 g + quinoa 60 g (dry) + roasted broccoli 200 g + 1 tbsp extra virgin olive oil + lemon.
- Snack: Greek yogurt (0% fat) 200 g + 1 tsp honey + 10 almonds.
- Dinner: Lentil and spinach stew (lentils 80 g dry, spinach 150 g, tomato, onion, garlic, cumin) + 1 small whole-grain pita.
- Optional: 1 small apple.

Day 2 (Tue)

- Breakfast: 3 eggs scrambled in 1 tsp olive oil + 2 slices whole-grain bread + 1 tomato sliced + spinach.
- Lunch: Chickpea salad (chickpeas 150 g cooked, cucumber, tomato, red onion, parsley, feta 30 g, olive oil 1 tbsp, lemon).
- Snack: 30 g pumpkin seeds + 1 pear.
- Dinner: Grilled chicken breast 150 g + bulgur 60 g (dry) + roasted zucchini and bell pepper + tzatziki (Greek yogurt + cucumber).

Day 3 (Wed)

- Breakfast: Smoothie (250 mL skim milk + 1 banana + 1 tbsp ground flaxseed + 25 g whey protein + handful spinach).
- Lunch: Whole-grain pasta 70 g (dry) with sardines 100 g, garlic, olive oil, cherry tomatoes, capers, parsley.
- Snack: 2 Brazil nuts + 1 orange.
- Dinner: Tofu stir-fry (tofu 150 g, broccoli, snow peas, mushrooms, 1 tsp sesame oil, soy sauce, ginger) over brown rice 50 g (dry).

Day 4 (Thu)

- Breakfast: Greek yogurt 250 g + oats 30 g + 100 g pomegranate arils + 10 walnut halves.
- Lunch: Tuna (light, canned in water) 120 g over mixed greens + white beans 100 g + olives + olive oil + balsamic.
- Snack: Carrot and cucumber sticks + 3 tbsp hummus.
- Dinner: Baked mackerel 150 g + roasted sweet potato 200 g + steamed green beans + olive oil and lemon.

Day 5 (Fri)

- Breakfast: 2 eggs + 1 slice whole-grain toast + 1/2 avocado + 1 tomato sliced + 1 cup green tea.
- Lunch: Lentil soup 400 mL + side salad with olive oil + 1 small whole-grain roll.
- Snack: 1 apple + 30 g almonds.
- Dinner: Grilled shrimp 150 g + farro 60 g (dry) + roasted asparagus + cherry tomatoes.

Day 6 (Sat)

- Breakfast: Overnight oats (oats 50 g, skim milk 200 mL, chia 1 tbsp, berries 100 g, walnut halves 10).

- Lunch: Grain bowl (quinoa 60 g dry, roasted vegetables, chickpeas 100 g, tahini-lemon dressing, pumpkin seeds 15 g).
- Snack: Cottage cheese 150 g + pineapple 100 g.
- Dinner: Lean beef sirloin 120 g (once-weekly red meat) + roasted Brussels sprouts + barley 60 g (dry).

Day 7 (Sun)

- Breakfast: Whole-grain pancakes (oats + egg + banana + cinnamon) + Greek yogurt + 100 g berries.
- Lunch: Whole-grain wrap with grilled chicken 120 g, hummus, spinach, tomato, cucumber.
- Snack: 2 squares 85% dark chocolate + 1 cup green tea.
- Dinner: Baked salmon 150 g + couscous (whole-wheat) 60 g + Mediterranean salad (tomato, cucumber, olives, feta, olive oil).

7. A practical 12-week protocol

7.1 Phases

Weeks 1-2: Establish

- Baseline: weight, waist circumference, BP, fasting glucose, HbA1c, lipids, total testosterone, SHBG, prolactin, TSH, vitamin D, ferritin; semen analysis if seeking conception.
- Replace sugar-sweetened beverages with water, unsweetened tea, or sparkling water.
- Add a daily 30 g serving of mixed nuts (include walnuts).
- Add oily fish 2x/week.

Weeks 3-6: Deficit and structure

- Move to ~500 kcal/day deficit using the meal plan.
- Walk 30 min/day, 5 days/week; begin 2x/week resistance training.
- Eliminate processed meat and trans fat sources.
- Sleep target: 7-8 hours, dark and cool room.

Weeks 7-12: Consolidate

- Re-evaluate weight; if loss < 2% of body weight by week 6, re-check portions and revise energy target.
- Repeat hormone panel and (if applicable) semen analysis at week 12. Spermatogenesis takes ~74 days, so the week-12 result reflects the diet started at week 0.
- Plan for maintenance: protein and produce stay high, energy intake returns to maintenance once goal weight is reached.

7.2 Considered supplements (only with clinician oversight)

- Vitamin D3 1000-2000 IU/day if 25-OH-D < 30 ng/mL.
- Folate 400-800 microg/day in food + supplement.
- Omega-3 (EPA+DHA) 1-2 g/day if fish intake is low.
- Zinc 15-25 mg/day; do not exceed 40 mg/day chronically.
- L-carnitine and coenzyme Q10 have RCT evidence in idiopathic asthenozoospermia; discuss with a urologist.
- Selenium: prefer 1-2 Brazil nuts/day rather than supplements; selenium toxicity is real.

7.3 Red flags to escalate to a clinician immediately

- BMI $\geq$ 35 with comorbidities (consider pharmacotherapy or bariatric evaluation in addition to diet).
- Total testosterone < 300 ng/dL on two morning samples.
- Azoospermia or severe oligozoospermia on semen analysis.
- Symptoms of obstructive sleep apnea, which independently depresses testosterone and worsens metabolic health.

8. Selected references

Citations are listed by topic. Articles can be retrieved by searching the title in PubMed (pubmed.ncbi.nlm.nih.gov). This list is illustrative, not exhaustive.

Mediterranean diet and male fertility

- Salas-Huetos A, et al. Adherence to the Mediterranean diet and semen quality: a systematic review with meta-analysis. (multiple cohort studies).
- Ricci E, et al. Mediterranean diet and outcomes of assisted reproduction. *Adv Nutr* (2025).
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- Influence of the Mediterranean diet on seminal quality: systematic review (PMC10902424, 2024).

Healthy dietary patterns and semen

- Salas-Huetos A, et al. The effect of healthy dietary patterns on male semen quality: systematic review and meta-analysis. *Asian J Androl* (2022).
- Salas-Huetos A, et al. The effect of nutrients and dietary supplements on sperm quality parameters: systematic review and meta-analysis of RCTs. *Adv Nutr* (2018).
- Diet and male fertility: nutrients and antioxidants on sperm energetic metabolism. *Antioxidants* (2022, PMC8910394).

Antioxidants, micronutrients and sperm

- Cochrane review: Antioxidants for male subfertility (Smits RM, et al., updated editions).
- Antioxidant supplementation on sperm DNA fragmentation and sperm parameters: systematic review and meta-analysis. *Turk J Urol* (2022).
- Antioxidant supplementation on male fertility - systematic review. *Antioxidants* (2023, PMC10135082).
- Network meta-analysis of antioxidants in idiopathic male infertility: L-carnitine, CoQ10, omega-3, selenium. *Front Endocrinol* (2022).
- Micronutrient supplements as antioxidants in improving sperm quality and reducing DNA fragmentation. *Basic Clin Androl* (2023).

Specific foods - walnuts, fish, pomegranate

- Robbins WA, et al. Walnuts improve semen quality in men consuming a Western-style diet: RCT. *Biol Reprod* (2012).
- Salas-Huetos A, et al. Effect of nut consumption on semen quality and functionality in healthy men. *Am J Clin Nutr* (2018).
- Safarinejad MR. Effect of omega-3 PUFA supplementation on semen profile and antioxidant capacity. *Andrologia* (2011).
- Pomegranate / galangal extract and motile sperm count: double-blinded RCT (PMC4190413).

Obesity, weight loss and male fertility

- BMI is associated with sperm quality and sex hormones in men: meta-analysis. *Front Endocrinol* (2025).
- Association between body mass index and semen quality: systematic review and meta-analysis (2024).
- Weight loss as therapeutic option to restore fertility in obese men. (*Andrology* review, 2024).
- Obesity and male fertility disorders. *Rev Endocr Metab Disord* (2024).
- The effect of obesity interventions on male fertility (*Hum Reprod Update* style, 2025).
- Effect of bariatric surgery on male infertility: updated meta-analysis (2025).

Diet patterns and weight loss in BMI 25+

- Esposito K, et al. Mediterranean diet and weight loss: meta-analysis of RCTs. *Metab Syndr Relat Disord* (2011).
- Mediterranean diet in the management and prevention of obesity. *Rev Endocr Metab Disord* (2023).
- Mediterranean diet and obesity-related disorders: evidence review (PMC9729142).
- Optimal diet strategies for weight loss and weight loss maintenance. *J Obes Metab Syndr* (2020).
- Effects of popular diets on anthropometric and cardiometabolic parameters: umbrella review of meta-analyses of RCTs.

- Mediterranean diet effects on insulin resistance and secretion in overweight/obese (PMC10648830).

Nutritional values per 100 g (data sources)

- USDA FoodData Central (fdc.nal.usda.gov).
- EuroFIR - European Food Information Resource Network.
- McCance and Widdowson's The Composition of Foods, 7th edition.

End of document. Compiled as an educational synthesis. Not a substitute for individualized medical advice.